

Definitions

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Propositionally Coherent Logical Inductors

Definition 1 (Propositionally coherent valuations).

- $\mathcal{S}_n$ is the finite set of prime formulas under consideration in day n .
We require $\mathcal{S}_n \subseteq \mathcal{S}_{n+1}$.
(They will be those that have appeared either in D_n , or in some T_m with $m \leq n$.)
- $\mathbb{W}^n$ is any boolean truth assignment to $\mathcal{S}_n$.
- $\mathcal{W}^n$ is the set of all $\mathbb{W}^n$. A PC-valuation $\mathbb{P}_n$ has support $\mathcal{W}^n$,
and furthermore satisfies $\sum_{\mathbb{W}^n \in \mathcal{W}^n} \mathbb{P}_n(\mathbb{W}^n) = 1$.
- An expressible feature is defined as always.
For its calculation, though, we define $\mathbb{P}_n(\phi) := \sum_{\mathbb{W}^n \models \phi} \mathbb{P}_n(\mathbb{W}^n)$
- Similarly, an n -trading strategy is defined as always,
but a share of ϕ is actually just the bundle with a share of each $\mathbb{W}^n$ with $\mathbb{W}^n \models \phi$.

Definition 2 (Accumulated wealth). Given an n -strategy history $T_{\leq n}$, and an n -belief history $\mathbb{P}_{\leq n}$, the *accumulated wealth* of T after round n is defined as

$$\mathcal{A}(T_{\leq n}, \mathbb{P}_{\leq n}, n) := \sum_{i \leq n} (\mathbb{P}_i(T_i[c]) + \sum_{\phi} T_{i-1}[\phi] \cdot \mathbb{P}_i(\phi))$$

where of course $\mathbb{P}_i(T_i[c])$ is calculated by just introducing the $\mathbb{P}_i$ prices (as defined above) into the expression for $T_i[c]$.

That is, every round the trader sells all their shares, receiving their new prices as payment for them, and then goes on to buy whatever shares. In that sense, each trader is trying to predict the next market state.

Definition 3 (Respecting a deductive process). A PC-market $\bar{\mathbb{P}}$ *respects* the deductive process $\bar{D}$ iff $\mathbb{P}_n(\phi) = 1$ for each $\phi \in D_n$. Equivalently, if all its worlds with non-zero probability satisfy ϕ .

Definition 4 (PC-Exploitation). A trader $\bar{T}$ is said to *PC-exploit* a valuation sequence $\bar{\mathbb{P}}$ if the set of values

$$\{\mathcal{A}(T_{\leq n}, \mathbb{P}_{\leq n}, n) \mid n \in \mathbb{N}^+\}$$

is bounded below but not above.

Definition 5 (PC-LIC). A market $\bar{\mathbb{P}}$ is said to satisfy the *Propositionally Coherent Logical Induction Criterion* relative to the deductive process $\bar{D}$ if

- $\bar{\mathbb{P}}$ respects $\bar{D}$, and
- No efficiently computable $\bar{T}$ PC-exploits $\bar{\mathbb{P}}$.

Conjecture 1 (Equivalence of Criteria). Define $\varphi_{\mathbb{W}^n}$ as the conjunction of all the prime formulas in $\mathcal{S}_n$, negated or non-negated according to their truth values assigned by $\mathbb{W}^n$. Given a PC-market $\bar{\mathbb{P}}$ respecting $\bar{D}$, define the regular market $\mathbb{P}'$ by $\mathbb{P}'_n(\varphi_{\mathbb{W}^n}) := \mathbb{P}_n(\mathbb{W}^n)$ (and same for their Boolean combinations). Then, $\bar{\mathbb{P}}$ satisfies the above Criterion relative to $\bar{D}$ iff $\mathbb{P}'$ satisfies the original Logical Induction Criterion relative to $\bar{D}$.

Sketch of Proof. Suppose $\bar{T}'$ exploits $\bar{\mathbb{P}}$. If some (infinite, and e.c.) world coherent with $\bar{D}$ (that is, $\mathbb{W}_\infty \in \mathcal{PC}(D_\infty)$) has limit probability 0 (that is, $\sum_{\mathbb{W}^n \models \mathbb{W}_\infty} \mathbb{P}_n(\mathbb{W}^n)$ tends to 0), then we can construct a $\bar{T}$ exploiting $\bar{\mathbb{P}}$ by always selling these probabilities. If that's not the case, then the worlds in which $\bar{T}'$ is obtaining unbounded returns also provide unbounded returns via the market prices. Also note that, since the accumulated wealth of $\bar{T}'$ is a weighted average of those in all possible propositionally coherent worlds, all the latter being bounded below implies the former being bounded below.

For the converse, suppose $\bar{T}$ PC-exploits $\bar{\mathbb{P}}$, that is, their returns from market prices are unbounded. Then, in particular, by considering the worlds corresponding to those market prices (which were possible worlds when $\bar{T}$ bought the shares, since otherwise they'd have been priced at 0), and considering that worlds only get more fine-grained with time, we see $\bar{T}$ also obtains unbounded returns across possible worlds. And for the lower bound, consider any sequence of possible worlds in which $\bar{T}$ makes unbounded losses. Then, since $\bar{T}$ doesn't make unbounded losses in market prices, these have a finite weight in its buyings, so we can eliminate them (by hard-coding this finite decision into $\bar{T}$)? $\square$

Conjecture 2 (Constructing a PCLI from an LI). Given $\bar{\mathbb{P}}$ a regular LI relative to $\bar{D}$, consider the PC-market $\bar{\mathbb{P}}$ defined as

$$\mathbb{P}_n(\mathbb{W}_i^n) := \frac{\mathbb{P}'_{n'}(\varphi_{\mathbb{W}_i^n})}{\sum_{\mathbb{W}^n} \mathbb{P}'_{n'}(\varphi_{\mathbb{W}^n})}$$

where n' is the smallest $m > n$ such that this fraction is well-defined. Then no efficiently computable $\bar{T}$ PC-exploits $\bar{\mathbb{P}}$.

Sketch of Proof. In the limit $\mathbb{P}'$ has an opinion about all these worlds (although maybe we need an assumption that $\mathcal{S}_n$ grows slowly enough, or a change in the definition to account for this). And in the limit the difference between the prices in both valuations is 0, since traders arbitrate coherence in $\mathbb{P}'$. $\square$

In any event, we can also directly construct a PC-LI.

Definition 6 (PC-Budgeter).

- If for some $m < n$ we have $\mathcal{A}(T_{\leq m}, \mathbb{P}_{\leq m}, m) \leq -b$, then $\text{PC-Budgeter}_n^{\bar{D}}(b, T_{\leq n}, \mathbb{P}_{\leq n-1}) := 0$.

- Otherwise,

$$\text{PC-Budgeter}_n^D(b, T_{\leq n}, \mathbb{P}_{\leq n-1}) := T_n \cdot \inf_{\mathbb{P}_n} \left[\max(1, \frac{-\mathbb{P}_n(T_n[c]) - \sum_{\phi} T_{n-1}[\phi] \cdot \mathbb{P}_n(\phi)}{b + \mathcal{A}(T_{\leq n-1}, \mathbb{P}_{\leq n-1}, n-1)})^{-1} \right]$$

So we are just calculating the current balance of the trader (if they had started with b), and then capping their trades so that they don't go negative even in the least favorable next market state.

This budgeter can be arbitrarily well-approximated, due to T_n being a Lipschitz function of the prices (I think). Nonetheless, maybe to avoid the complication I want a different definition.

Definition 7 (PC-TradingFirm). Exactly as in the paper, but with PC-Budgeter.

Theorem 1 (PC-TradingFirm dominance). If an e.c. trader PC-exploits $\bar{\mathbb{P}}$, then so does PC – TradingFirm.

Theorem 2 (Fixed point lemma). Given a T_n , there is a PC-valuation $\mathbb{P}_n$ over $\mathcal{W}^n$ such that, for any $\mathbb{P}_{n+1}$, T_n doesn't make a profit.

Sketch of Proof. The tricky bit is ensuring that the function goes from the simplex to the simplex. That is, that it preserves both $\sum_{\mathbb{W} \in \mathcal{W}^n} \mathbb{P}(\mathbb{W}) = 1$ and $\mathbb{P}(\mathbb{W}) \in [0, 1]$.¹ Define the benefit obtained by T_n in world $\mathbb{W}$ given prices $\mathbb{P}$ as

$$B(\mathbb{W}) := T_n(\mathbb{P})[\mathbb{W}] \cdot 1 + T_n(\mathbb{P})[c] = T_n(\mathbb{P})[\mathbb{W}] \cdot 1 - \sum_{\mathbb{W} \in \mathcal{W}^n} T_n(\mathbb{P})[\mathbb{W}] \cdot \mathbb{P}(\mathbb{W})$$

(where we've ignored some dependencies for ease of notation).

We want all of the $B(\mathbb{W})$ to be ≤ 0 . Indeed, this is necessary and sufficient to ensure that no market state in the next round would yield a benefit to the trader in accumulated wealth. I want to define something like

$$\text{fix}(\mathbb{P})(\mathbb{W}) := \mathbb{P}(\mathbb{W}) + B(\mathbb{W}) \cdot \mathbb{P}(\mathbb{W}) \cdot \delta(\mathbb{P})$$

where $\delta(\mathbb{P})$ is a bounding term ensuring we stay within the simplex:

$$\delta(\mathbb{P}) := \min(1, \max \{ r \mid \forall \mathbb{W} (\mathbb{P}(\mathbb{W}) + B(\mathbb{W}) \cdot \mathbb{P}(\mathbb{W}) \cdot r \in [0, 1]) \})$$

We also get $\sum_{\mathbb{W} \in \mathcal{W}^n} \text{fix}(\mathbb{P})(\mathbb{W}) = 1$ because $\sum_{\mathbb{W} \in \mathcal{W}^n} B(\mathbb{W}) \cdot \mathbb{P}(\mathbb{W}) = 0$.

Now given a fixed point $\mathbb{P}^{\text{fix}}$ of this function, either $B(\mathbb{W}) = 0$ (in which case we are good), or $\delta(\mathbb{P}^{\text{fix}}) = 0$ (in which case we get a contradiction), or $\mathbb{P}^{\text{fix}}(\mathbb{W}) = 0$ (in which case... this breaks). I've tried similar functional forms, by addressing more and more counterexamples, but each has failed somewhere.

I've also tried ideas like

$$\text{fix}(\mathbb{P})(\mathbb{W}) := \frac{\mathbb{P}(\mathbb{W}) + (B(\mathbb{W}) - \min_{\mathbb{W}' \in \mathcal{W}^n} \{B(\mathbb{W}')\})}{\sum_{\mathbb{W}'' \in \mathcal{W}^n} \mathbb{P}(\mathbb{W}'') + (B(\mathbb{W}'') - \min_{\mathbb{W}' \in \mathcal{W}^n} \{B(\mathbb{W}')\})}$$

This sums to 1 and remains in $[0, 1]$. Given $\mathbb{P}^{\text{fix}}$, consider a $\mathbb{W}$ with $B(\mathbb{W}) = \min_{\mathbb{W}' \in \mathcal{W}^n} \{B(\mathbb{W}')\}$. Then either the below summatory is 1 (in which case we're done), or $\mathbb{P}^{\text{fix}}(\mathbb{W}) = 0$. In the latter case, I haven't been able to prove a contradiction by assuming $B(\mathbb{W}') > 0$ for some $\mathbb{W}'$. $\square$

Definition 8 (PC-MarketMaker). Approximate the fixed point.

¹Indeed, we cannot just ensure that the trader only buys at the price of 1 or sells at the price of 0, as in the original paper. Consider a trader that buys a share of $\mathbb{W}_1^n$ and one of $\mathbb{W}_2^n$. Then we can't put price 1 in both. What we can do is 0.5 and 0.5 (or any other mixture), which amounts to putting price 1 in $\mathbb{W}_1^n \vee \mathbb{W}_2^n$, which is what the trader is really buying.

LUV Coherent Logical Inductors

The same as above but with worlds including LUVs syntactically, as I had already done. That is, when a world believes X to be a $[0, 1]$ -LUV, it must also assign its value with precision $1/n$, and we perform several consistency checks on each of these worlds (eliminating those that don't satisfy them). We do get the complication that some worlds are more granular than others, but I don't think this will mess up the fix-point.

Omni-traders

Consider just infinite n -strategies.

Open problem: Do they always have a fixed point? (Disregarding computability)

I'd like a σ -algebra over possible market states, with sets of the kind

$$\{\mathcal{Q} \mid l_i \leq \mathcal{Q}(\varphi_i) \leq u_i \text{ for } i \in I\}$$

or

$$\{\mathcal{Q} \mid l_i \leq \mathcal{Q}(\varphi_i) \leq u_i \text{ for } i \in I, \text{ and } \mathcal{Q}(\varphi) \in [0.5 - \varepsilon, 0.5 + \varepsilon] \text{ for all other } \varphi\}$$

This is fundamentally unworkable given computability, because of closure under countable union. There's another way in which the LI paradigm is not natural for this: in it, when we don't yet have an opinion on φ (it's outside the Support) we go $\mathbb{P}_n(\varphi) = 0$, while here we go $\mathbb{P}_n(\varphi) \approx 0.5$.

But whatever, let's ditch computability of this kind. Can I make this work in a purely Bayesian framework? Consider each n -trader is a Bayesian hypothesis $T_n : \mathcal{S} \rightarrow [0, 1] \cap \mathbb{Q}$, and I Bayesian update on them. Well, if they are arbitrary functions, then updating on them gives me no information (how do I weigh them?). If they are computable functions (weighed by simplicity, or something), then I will just converge on the one generating all theorems. If I impose more efficient computability than that, then we get non-realizability: none of our hypotheses correspond to the process generating observations.

I think this all hints at a more general problem:

How do we provide non-infinitesimal incentive to get large sentences (or market-states containing them) right? Because any reward function will be simplicity-biased, and thus traders will learn to ignore large enough sentences.

We might be able to overfit this assessment to the finite $\mathbb{Q}_n$, but actually $\mathbb{Q}_n$ has to be infinite, so we have the same problem.

This seems deeply related to correct allocation of computation. In a realistic situation, the trader has to decide on which sentences to center its computation. In this situation, the computation spent in each sentence is independent. We might want to constrain this (the total computation has to sum up to something), but this will be just usual traders.

Main thing we're lacking (after today's call): Coherence constraints for large beliefs from small beliefs, of the decision-relevant kind.

Traders with memory

This is motivated by the original paper's unattained Decision Rationality Desiderata, and trying to predict Ω .

In a regular LI, traders can consult past market prices, but not their past shares. What if we allow this? (Or analogously, allow the trader some access to memory.) Then new trading strategies become possible, like thinking for many rounds about a complicated sentence, and then betting on it. Indeed, if at each round the trader has $\text{poly}(n)$ computation, by the time the trader stops thinking they will have used $\text{poly}(n)!$ computation, which is more than any trader can do only in round n .

I'm not sure about the complexity of this, I think it will be neither a polynomial nor an exponential LI. Maybe it's just a factorial LI, since in the limit, at each round, new traders who have been thinking for all past rounds will "awaken" and bet. But this is not clear, since these traders might also appear more sparsely?

Vetoing traders

Consider $\text{LIA}(\sigma)$, where $\sigma \in \{0, 1\}^*$. This works exactly like LIA , except that the traders with index i satisfying $\sigma(i) = 1$ get eliminated (from the start, forever).

Consider now LIA facing a certain decision tree. Also, externally, we use a different inductor $\mathbb{O}$ observing LIA 's actions in the tree, and Ω 's responses. Call $U_n(\text{LIA}(\sigma))$ the utility that $\text{LIA}(\sigma)$ would have accumulated in this tree by round n (notice this isn't a counterfactual, just a different mathematical statement). In round n , we calculate $\mathbb{E}_n^{\mathbb{O}}(U_n(\text{LIA}(\sigma)))$ for a bunch of values of σ (for example, all those affecting the finite amount of traders in LIA_n). If this value is greater than $U_n(\text{LIA})$, that means our agent would have been better off ignoring those traders with $\sigma(i) = 1$. Thus, we punish those traders in some way, so that they steer behavior less in the future. Probably the best way to implement this punishment is by making them lose money proportional to the utility difference (and possibly n). We could also more directly scale down their trades, without making them go bankrupt.

This probably needs the following refinement: Instead of only considering finite $\sigma \in \{0, 1\}^*$, we consider all infinite σ describable by an e.c. algorithm (and introduce more of these algorithms as n grows). This way insights of the kind "all traders that bet on Don't Pay are bad" can be extrapolated into the future, instead of passively waiting for more of these traders to appear to shoot them down.

Note: Of course, we could have even richer counterfactuals of the kind "we have σ up until round n , then σ' , etc.". Not yet clear how to take advantage of this information though (how to implement them in punishment).